

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed Python scripts to automate the generation of weekly operational reports, reducing manual effort by approximately 15 hours per week for the finance operations team.
- Designed and implemented REST APIs using **Java Spring Boot** to facilitate data exchange between enterprise systems, improving data consistency across interconnected platforms.
- Managed production incidents for critical enterprise applications, responding to P1/P2 alerts within a **15-minute SLA** and performing root cause analysis to prevent recurrence.
- Containerized microservices using **Docker** and deployed them to **AWS EC2** instances, enabling more consistent and reliable application deployments.
- Authored **Python** scripts for data extraction and transformation from PostgreSQL databases, supporting the creation of analytical datasets for business intelligence reporting.
- Participated in Agile ceremonies, including sprint planning and daily stand-ups, contributing to the timely delivery of features and bug fixes within a **Scrum framework**.
- Maintained and optimized **SQL** queries across PostgreSQL and Oracle databases for data validation and ad-hoc reporting, ensuring data integrity for financial operations.

SKILLS

Languages – Python, Java, SQL, R, C++, C

Frameworks & Libraries – Spring Boot, pandas, NumPy, scikit-learn, PyTorch, TensorFlow, LangChain, HuggingFace, FastAPI, Flask

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, Jenkins, Git, Helm, Perforce

Databases – PostgreSQL, MySQL, Oracle, Vector Databases

AI/ML – Machine Learning, LLMs, Prompt Engineering, ML Ops, AI Ops, Model Lifecycle, Model Deployment, Model Monitoring, GenAI, Agentic AI

Operating Systems – Linux, Windows

Networking – TCP/IP, UDP

PROJECTS

Automated Data Pipeline for AI Model Training

- Engineered an end-to-end Python data pipeline on **AWS Lambda** to ingest and preprocess ~50GB of raw transaction data daily, preparing features for downstream ML model training.
- Developed **FastAPI** microservices to serve model predictions with p95 latency under 200ms, integrated with existing enterprise systems and handling approximately 500 requests per second.
- Implemented data validation checks within the pipeline to ensure the quality and consistency of input features, reducing model retraining failures by **30%**.

Customer-Facing AI Agent Development Framework

- Built a modular Python framework using **LangChain** and **RAG** principles to enable rapid development of AI agents for enterprise customers, supporting custom data ingestion and prompt engineering.
- Designed and implemented a **vector database** integration layer (using ChromaDB) for efficient retrieval of contextually relevant information, enhancing agent response accuracy by an estimated **20%**.
- Collaborated with a small engineering team to create example agent workflows and documentation, facilitating faster onboarding for two pilot customer engagements.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)